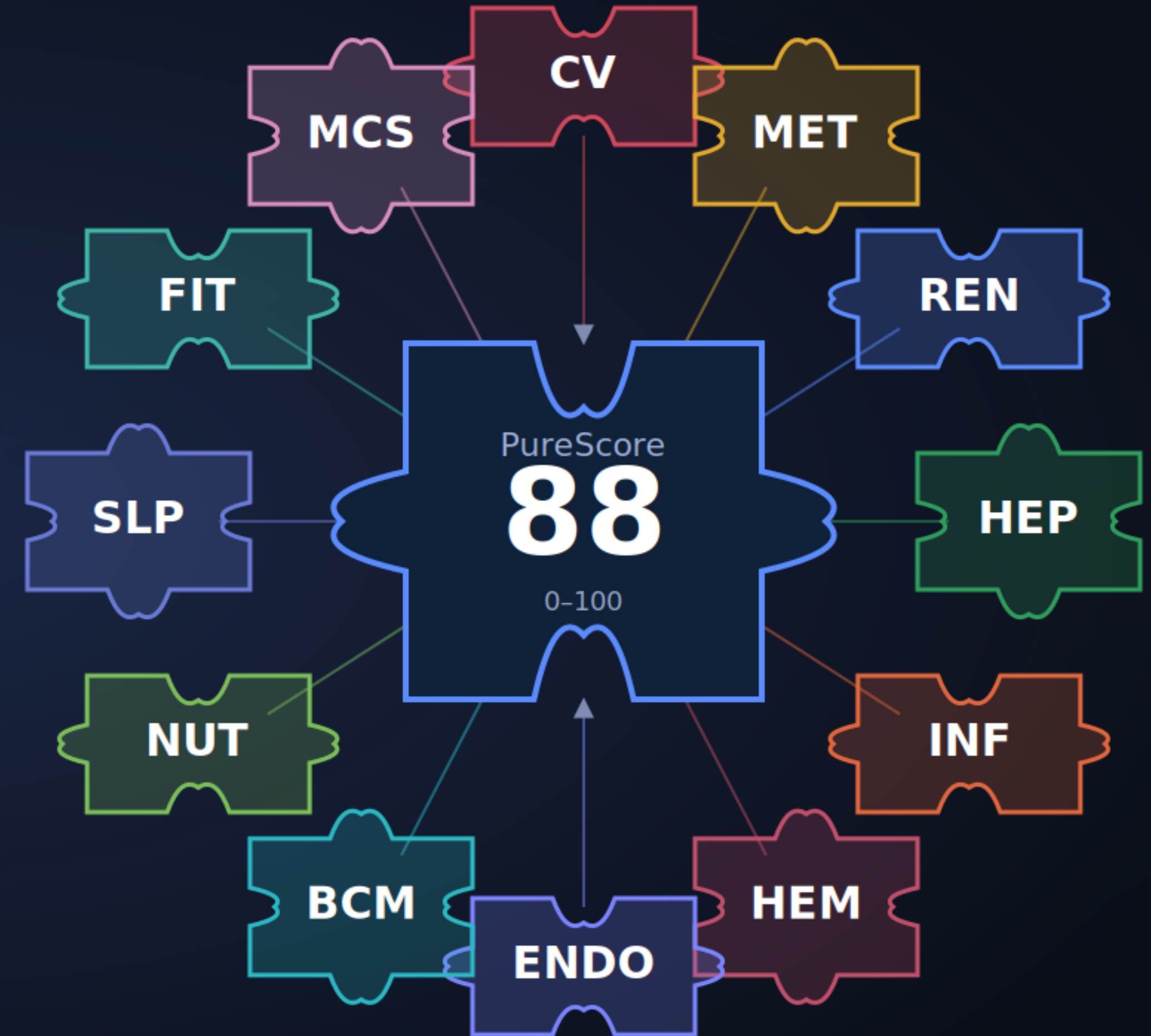


# PureScore

## The AI doctor that compounds.

A wellness-grade, clinician-in-the-loop health-intelligence engine: every patient is scored, watched, and coached — and the system gets sharper with every measurement it sees.

- **12 health pillars → one trusted score**
- **Catches disease while it is still incubating**
- **Localized for the UAE, governed by evidence**
- **Self-improving via a data + reinforcement loop**



# Why health scores fail

A single number, naively built, is confidently wrong.

- **Context blindness**

The same value means opposite things: resting HR 50 is elite in an athlete, dangerous on a  $\beta$ -bl

- **Too late**

Level-based thresholds only fire once disease has crossed a line — no early warning.

- **Over-confident & opaque**

One number, no error bars, no 'says who?' — clinicians can't trust or audit it.

- **Not personal, not local**

Western adult averages mis-score the UAE's South-Asian-majority population.

## The Babylon lesson

A celebrated 'AI doctor' over-promised, under-validated, and lost clinical trust.

“ No claim ships ahead  
of its evidence. ”

PureScore is built as the  
**deliberate antidote.**

# Twelve pieces. One picture.

Each health domain is a puzzle piece that locks into the whole.

- **The puzzle**

12 pillars (heart, metabolic, kidney, liver...) interlock — a gap in one weakens the whole.

- **The score**

Worst-sensitive: a single life-threatening value can't be averaged away by good numbers elsewhere.

- **The memory**

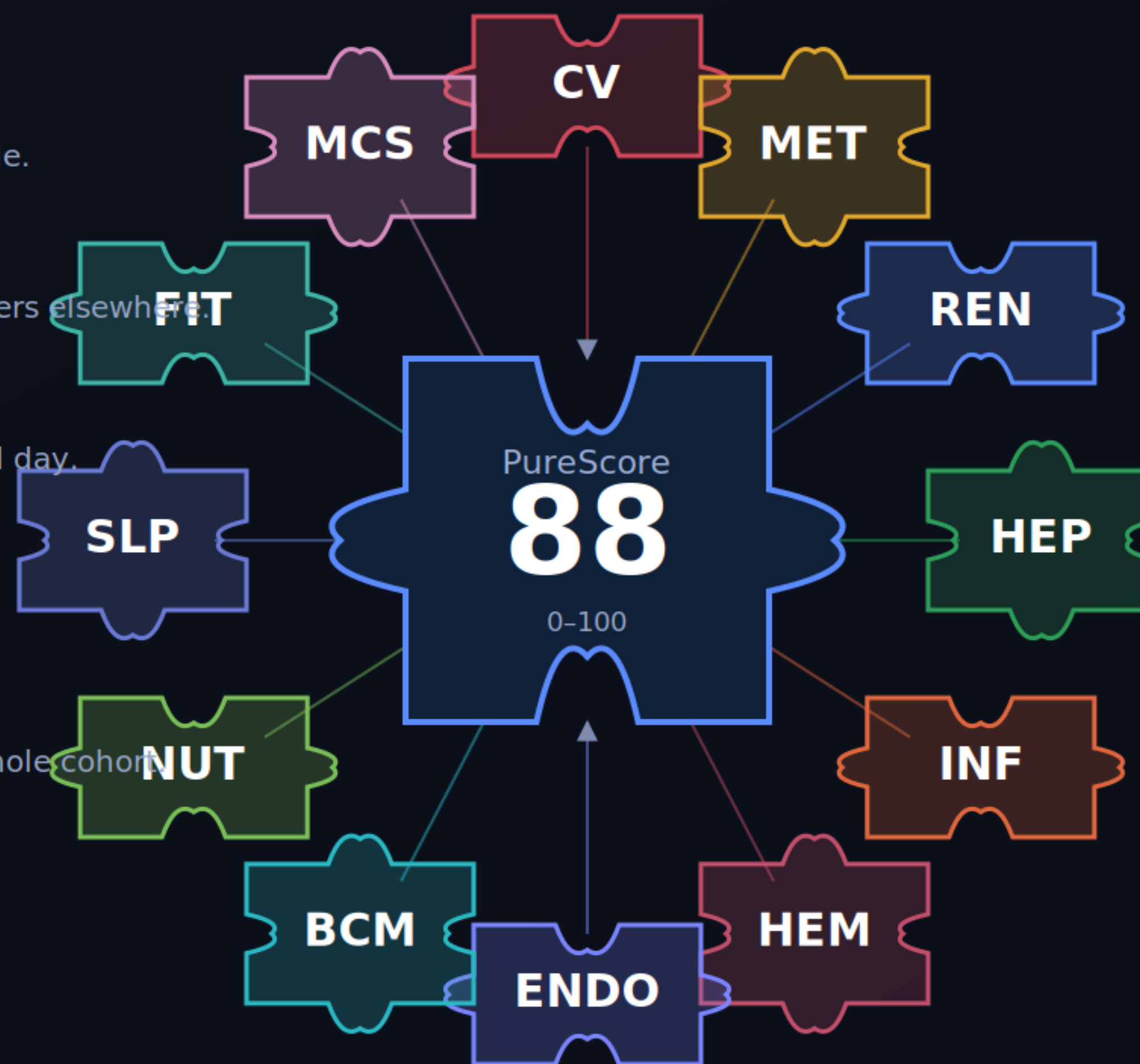
MONIAC 'reservoirs' store accumulated burden — a chronic debt shows even on a good day.

- **The trust layer**

The score ships with its own uncertainty, direction, and citations.

- **The loop**

Personal baselines + outcomes feed back — the engine learns each patient and the whole cohort.





# The patient at the centre

Every available signal, scored against the best reference — with honest coverage.

- **Use the patient's own data when present**

Fall back to literature/cohort references when absent — a score is always produced.

- **Tiered & coverage-aware**

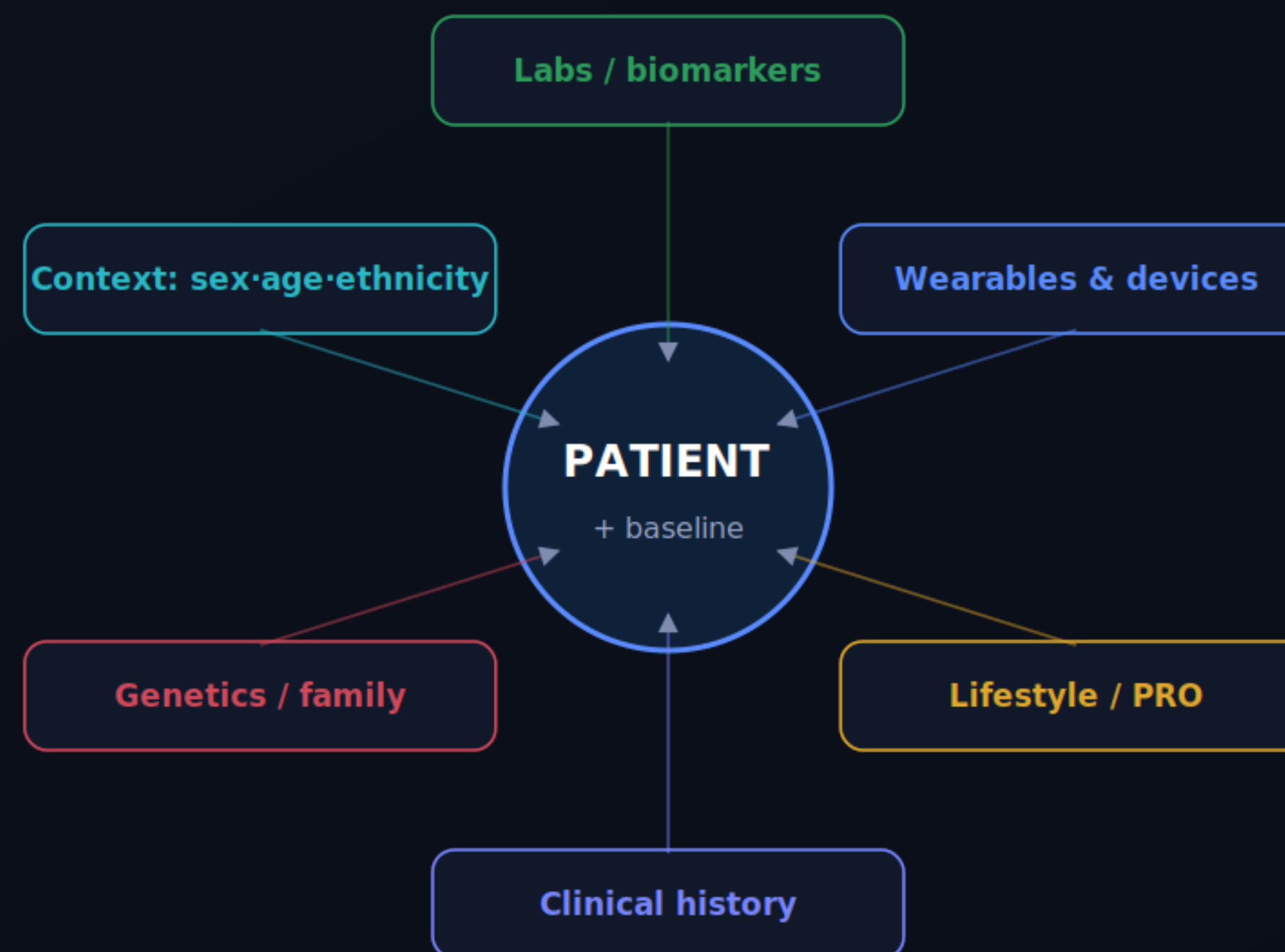
Each input has a tier and recency; missing data lowers Confidence, never fabricates it.

- **Organ-inventory & context aware**

Only scores what applies; captures sex-at-birth, hormones, life-stage, ethnicity.

- **Privacy by design**

Sensitive data (genetics, pregnancy) is minimized, gated, and never used to price or deny.



# The 12-pillar puzzle → PureScore

Marker → pillar → score, with a critical cascade and reservoir memory.

- **Marker risk  $r$**

Each value mapped to 0→1 risk against a green / yellow / red band (clinical + cohort).

- **Pillar risk  $R_k$**

$\gamma$ -power-mean of its markers — the worst marker dominates (fix the binding constraint).

- **PureScore**

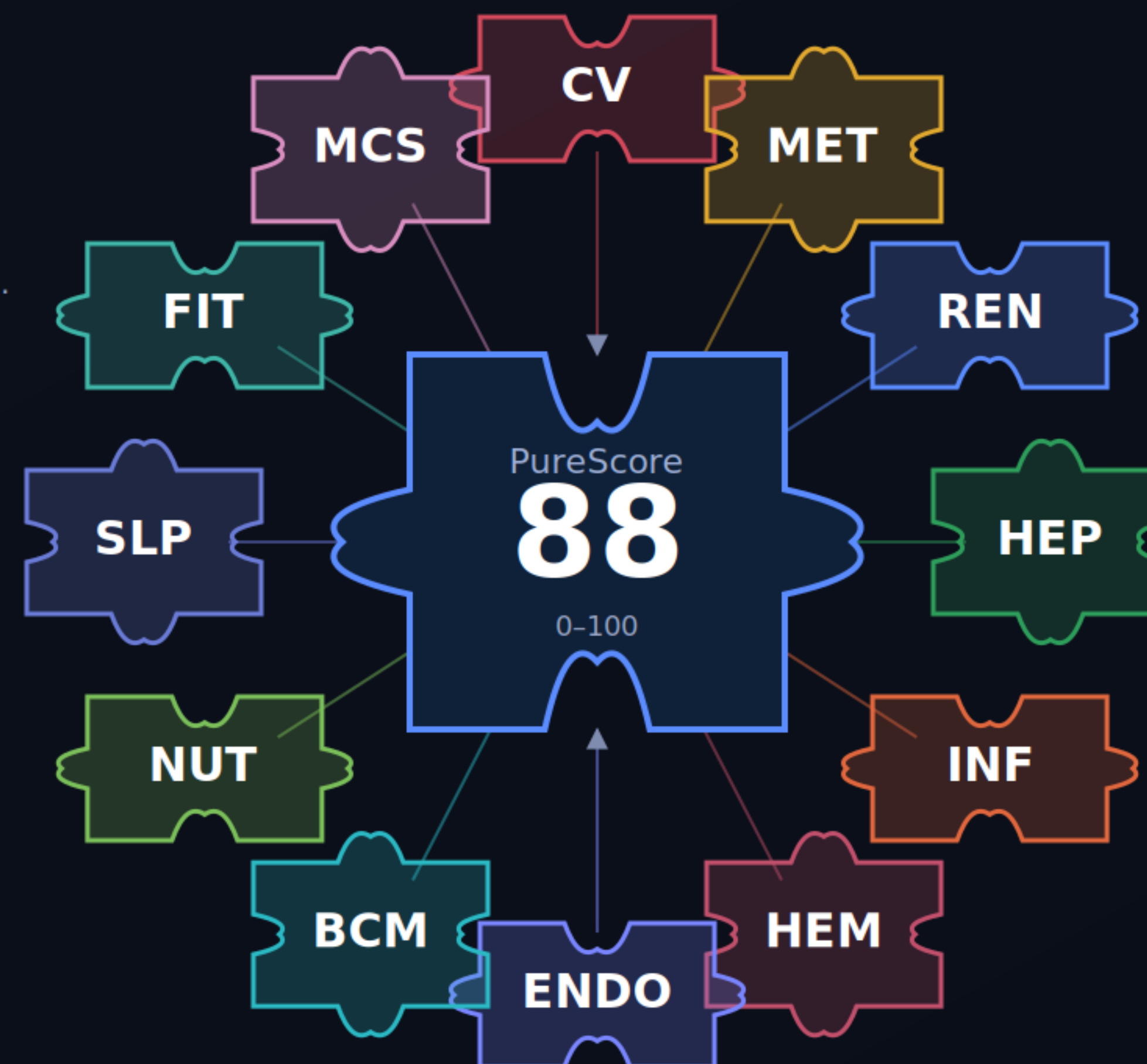
$100 \cdot (1 - \delta\text{-power-mean of pillars})$  — worst-sensitive across domains.

- **Critical cascade**

One red critical marker caps the score at 40 → cannot be averaged away.

- **Reservoir memory**

Sleep debt, atherogenic/glycemic/inflammatory burden inject cross-pillar latency.



$$\text{PureScore} = 100 \cdot (1 - R_{\text{total}})$$

$$R_{\text{total}} = \left( \frac{\sum W_k \cdot R_k^{\delta}}{\sum W_k} \right)^{1/\delta} \quad [\delta=2, \text{worst-sensitive}]$$

if any critical red → PureScore ≤ 40 (emergency escalation)

# A score with error bars

The number never travels alone — it carries its own honesty.

## Confidence

is the data trustworthy?

## Data sufficiency

measured vs imputed

## Criticality

independent red-flag badge

## Trajectory

improving / worsening

## Early-warning

Watch · Advisory · Alert

## Representativeness

does the cohort fit you?

## Skew / robustness

heavy-tailed handling

## Volatility

measurement stability

## Modifiability

how much is in your control

A '72 from fresh labs' and a '72 from imputation' no longer look identical.



# Catch it while it's incubating

Stop waiting for thresholds — watch personal baselines and motion.

- **Personal baseline (z-score)**

Your resting HR 52→60 matters even though both are 'green'.

- **Within-green drift & forecast**

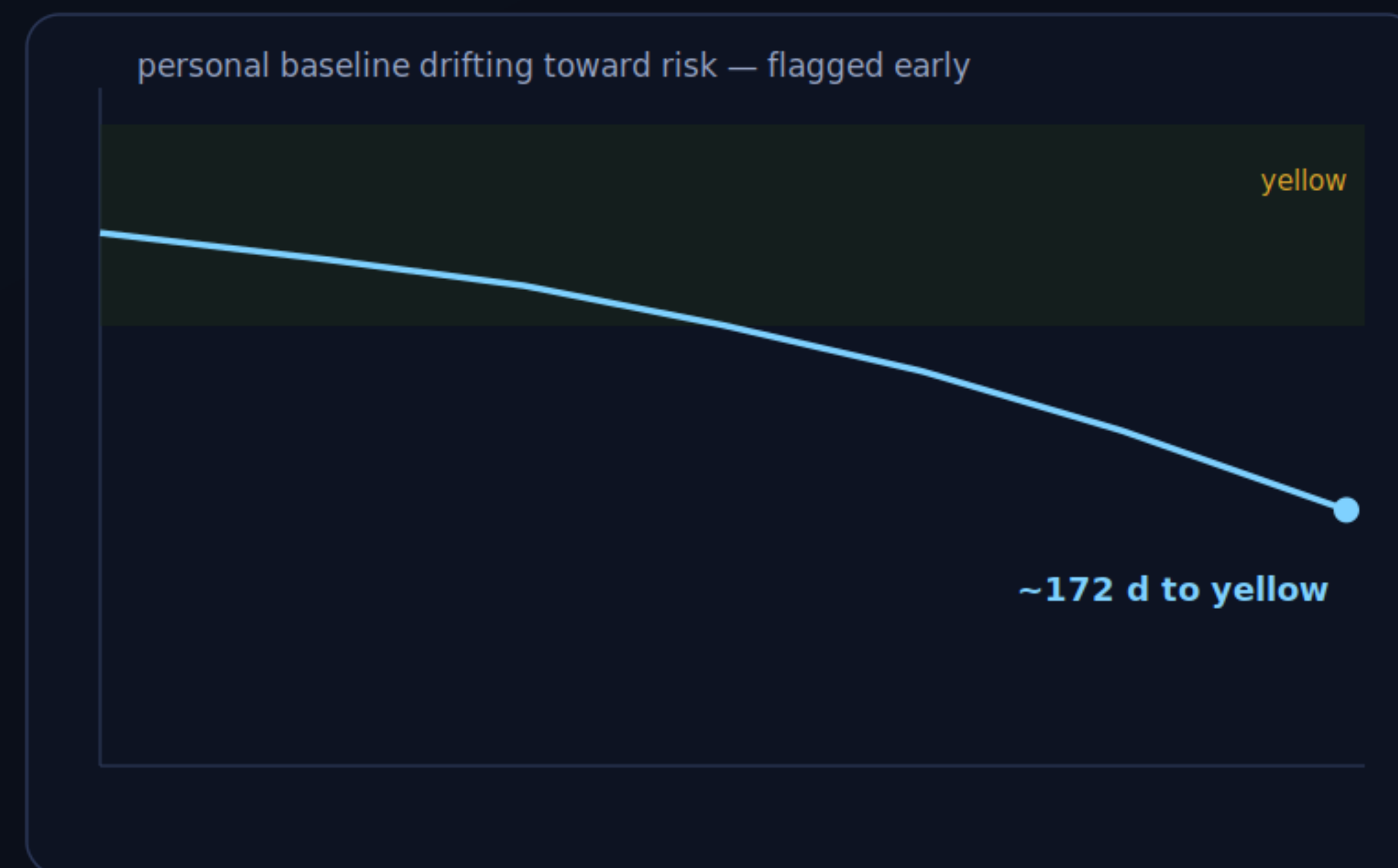
Projects days-to-threshold before a band is ever crossed.

- **Multivariate anomaly**

Many small simultaneous shifts — early sepsis, decompensation — caught together.

- **Tiered & buffered**

Watch (internal) → Advisory (gentle) → Alert (clinician). No alarm fatigue.





# Same value, right meaning

The engine reads each marker through the patient's real context.

## On medication

Controlled = 'MANAGED', not cured;  $\beta$ -blocker stops corrupting fitness.

## Sex & hormones

Hgb follows the hormonal milieu — correct for HRT, pregnancy, menopause.

## Age & life-stage

J-curve for frail elders; pregnancy trimester frames; never under-treat.

## Ethnicity (UAE)

WHO South-Asian BMI/waist cut-points; race-free eGFR — context, never penalty.

## Ramadan

Med-timing & hydration; absolute break-the-fast safety rule.



# The diesel-engine ignition

Cold-crank on guidelines, warm up on cohorts, run on the patient.

- **Every band cites its evidence**

42 registry entries: ADA, KDIGO, ESC, WHO, IDF + UAE bodies (MoHAP, DHA, IDF-DAR).

- **Weights shift with data**

Empirical-Bayes: the more the patient measures, the more personal the reference.

- **Safety never leaves the crank**

Acute-danger anchors stay absolute regardless of personalization.

- **Kept current by a crawler**

Guideline changes are detected, staged, and human-approved before anything moves.

- **Auditable**

A clinician can ask any number: 'says who, from when?' — and get the citation.

## CRANK Clinical guideline anchor

cold start · no patient data

$\theta \approx 0.62$

## WARM-UP Real-world cohort range

some data · cohort percentile

$\theta \approx 0.50$

## RUNNING Personal baseline z-score

established · self-sustaining

$\theta \approx 0.20$

# Five easiest wins, attributed

Impact-ranked daily actions — honest about what each is worth.

- **Ranked by ease × impact**

$U = [\Delta\text{PureScore} \cdot \text{evidence}] \cdot \text{adherence} \cdot \text{ease}$  — high-impact actions you won't do score zero.

- **Exact, honest impact**

$\Delta\text{PureScore}$  recomputed through the engine — diminishing returns in green, no false hope.

- **Modifiability-gated**

Lifestyle can't move a genetic marker (FH ApoB) → it routes to a clinician instead.

- **Safe & local**

CKD protein cap, pregnancy/lactation caveats, Ramadan timing baked in.

**Cut refined carbs**

MET · effort ●●○○○

**+5.7****2× resistance training**

BCM · effort ●●●○○

**+3.4****Protect 7.5-8 h sleep**

SLP · effort ●●○○○

**+1.6****+1,500 steps / day**

FIT · effort ●○○○○

**+1.4****Book clinician review**

CLINICAL · genetic ApoB

**refer**

# Built around the clinician

PureScore orchestrates the entities; humans hold the critical decisions.

- **Clinician-in-the-loop**

Wellness-grade by design: every red/critical escalates to a human — no autonomous diagnosis.

- **Closed care loops**

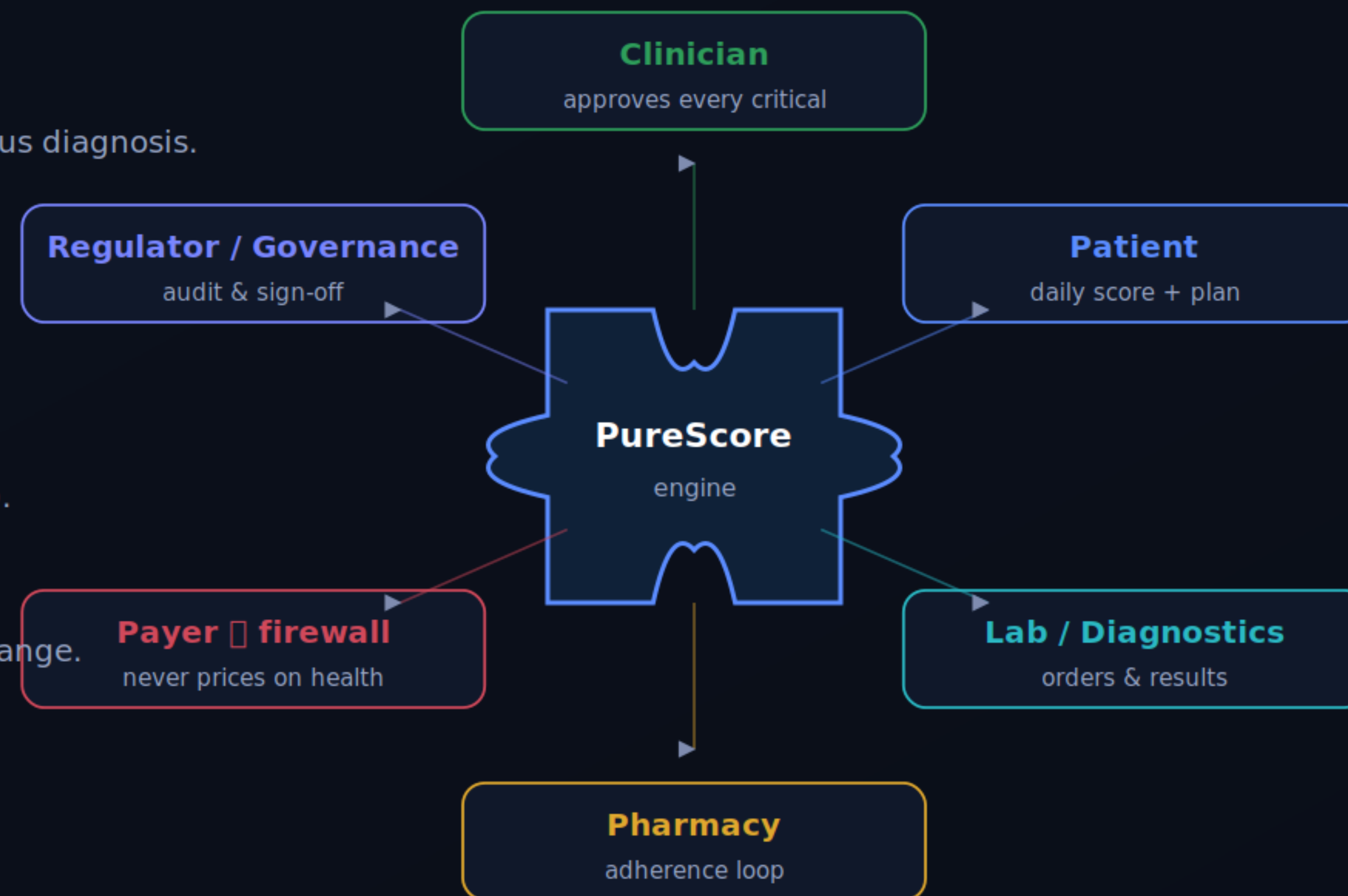
Lab gaps → MEASURE nudges; prescriptions → adherence; escalation → follow-up.

- **Payer firewall**

Health data & protected attributes can never worsen access or pricing (Doc 10/11).

- **Governable & auditable**

Evidence registry + model-version gates + human approval on every guideline change.

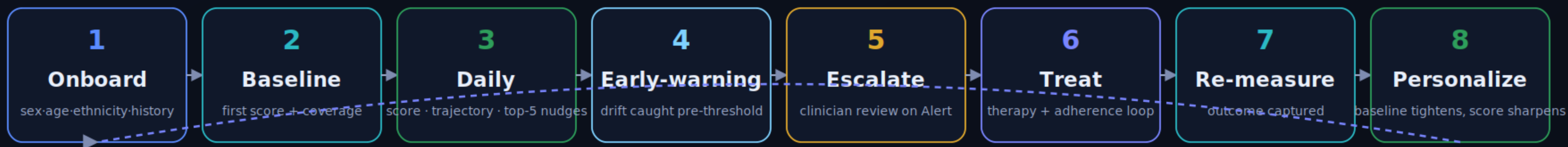




# From first measurement to lifelong partner

One continuous loop across every entity.

- **Continuous, not episodic — care between visits, not just at them.**



...and the loop repeats — every cycle the engine knows the patient better.

# How the AI doctor keeps improving

Data + reinforcement loop → compounding intelligence.

- **Reinforcement signal**

Realized  $\Delta$ PureScore vs predicted trains adherence & impact models per patient.

- **Empirical-Bayes shrinkage**

Personal evidence overrides the cohort prior as it accrues — no cold-start cliff.

- **Cohort learning**

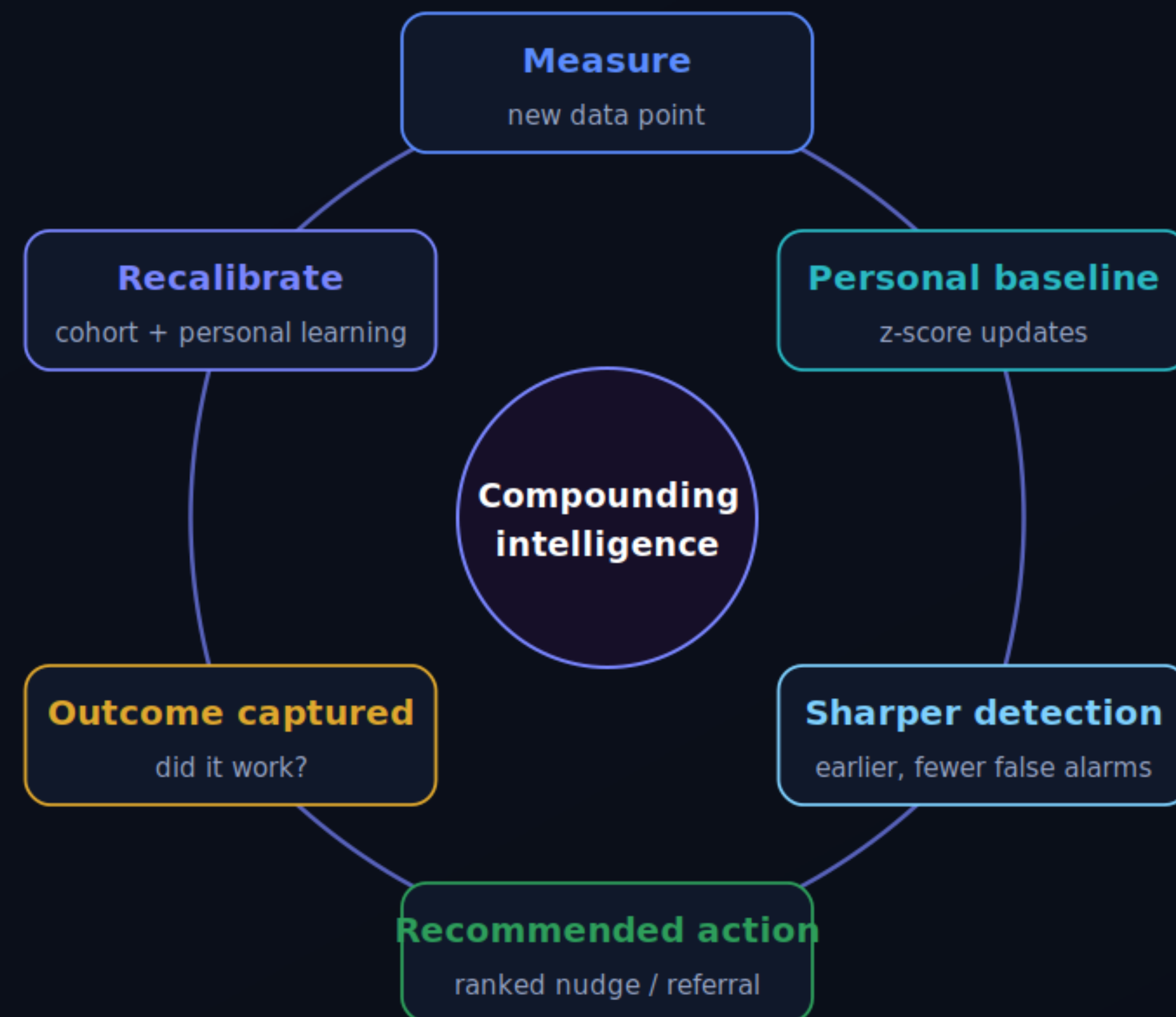
Each patient's outcomes recalibrate bands for similar patients (privacy-preserving).

- **Governed updates**

Every recalibration is a versioned, validated, human-approved release.

- **The moat**

More data → better personalization → better outcomes → more data.



# Earned, not asserted

The discipline that lets a clinician rely on it.

## ✓ Calibration gate

recalibrated ECE  $\leq 0.05$  before any risk read

## ✓ Early-warning gate

PPV at realistic prevalence + alarm budget

## ✓ Equity firewall

no protected attribute worsens score / price / access

## ✓ Fairness gate

parity across sex, ethnicity, life-stage (ratio  $\geq 0.80$ )

## ✓ Human-in-the-loop

every critical & every guideline change approved

## ✓ Evidence registry + crawler

citations kept current under review gate

Executable validation harness runs these gates today on synthetic data — verdict: NO-SHIP until real-data proof.



# From design to evidence

A credible path — honest about what remains.



The deliverable today is a defensible design — its value is unlocked by the data + reinforcement loop over time.

# Cardiovascular

Heart & arteries — the world's leading cause of death.

---

- **What it measures**

- Systolic blood pressure
- ApoB / atherogenic lipids
- Resting heart rate

- **Reservoir / coupling**

Atherogenic burden ( $\lambda \rightarrow 0$ : a near-permanent, decades-long memory).

- **Why it matters**

A statin-controlled ApoB reads MANAGED — credited, but never 'cured'.



# Metabolic

Blood sugar & insulin handling — the UAE's defining burden.

---

- **What it measures**

- HbA1c
- Fasting glucose
- Waist circumference

- **Reservoir / coupling**

Glycemic burden — feeds CV, renal, hepatic risk.

- **Why it matters**

WHO South-Asian waist cut-points catch risk a normal BMI hides.





# Renal

Kidney filtration & damage — silent until late.

---

- **What it measures**

- eGFR (race-free CKD-EPI 2021)
- UACR (albuminuria)
- Potassium

- **Reservoir / coupling**

Couples to glycemic & atherogenic burden.

- **Why it matters**

UACR rises before eGFR falls — a true leading indicator.





# Hepatic

Liver health & fibrosis (MASLD/NAFLD).

---

- **What it measures**

- ALT
- FIB-4 score
- Bilirubin

- **Reservoir / coupling**

Hepatic-fat burden, linked to metabolic load.

- **Why it matters**

FIB-4 stratifies fibrosis risk early, non-invasively.



# Inflammation

Systemic inflammatory load — a cross-cutting amplifier.

---

- **What it measures**

- hsCRP
- White cell count
- Albumin

- **Reservoir / coupling**

Inflammatory load — couples into CV, MET, hepatic, mood.

- **Why it matters**

Heavy-tailed CRP is handled on a log scale to avoid false alarms.



# Hematologic

Blood & oxygen carriage.

---

- **What it measures**

- Hemoglobin
- SpO<sub>2</sub>
- Platelets

- **Reservoir / coupling**

Iron/oxygen reserve; hemoglobinopathy context (regional).

- **Why it matters**

Hgb reference follows the hormonal milieu, not the birth certificate.





# Endocrine

Hormones, thyroid & the reproductive axis.

---

- **What it measures**

- TSH
- Cortisol (AM)
- Sex hormones (T / E<sub>2</sub>)

- **Reservoir / coupling**

Allostatic/stress reservoir coupling.

- **Why it matters**

Cycle, pregnancy, menopause, andropause & GAHT all reframe the bands.





# Body Comp / MSK

Muscle, fat distribution & bone.

- **What it measures**

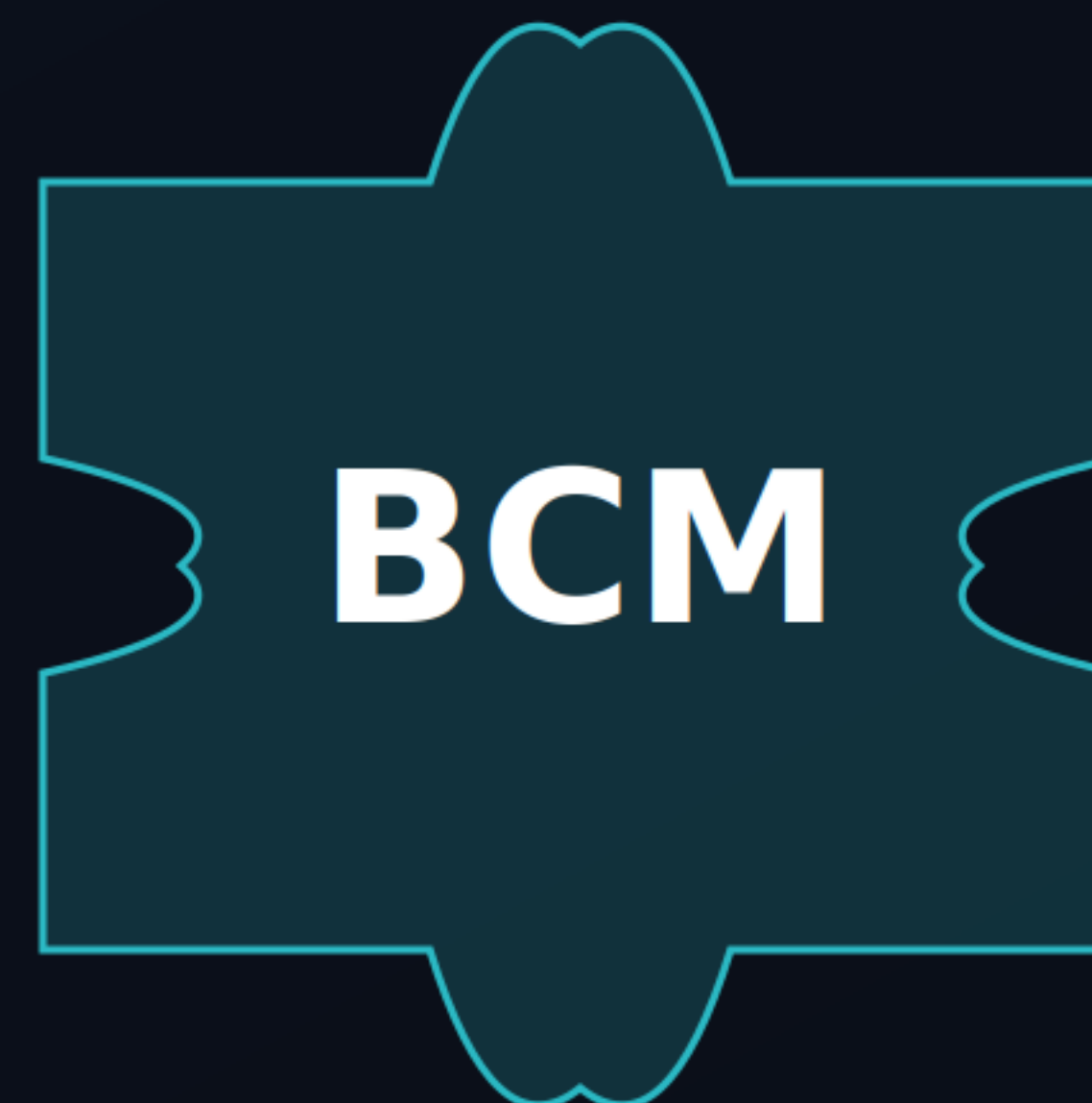
- BMI (ethnicity-aware)
- Grip strength
- BMD T-score

- **Reservoir / coupling**

Muscle reserve — protects metabolism & independence.

- **Why it matters**

Grip & VO<sub>2</sub>max surface sarcopenic frailty before a fall.



# Nutrition

Micronutrient sufficiency.

---

- **What it measures**

- Vitamin D
- Omega-3 index
- Vitamin B12

- **Reservoir / coupling**

Feeds inflammation & bone reservoirs.

- **Why it matters**

Vitamin-D deficiency is near-universal in the UAE — routine repletion.



# Sleep

Sleep quantity, quality & rhythm.

---

- **What it measures**

- Duration
- Efficiency
- Regularity

- **Reservoir / coupling**

Sleep debt (drains in days; the biggest hidden drag).

- **Why it matters**

One sleep action improves sleep, metabolic, mood & cardiovascular at once.





# Fitness

Cardiorespiratory capacity & activity.

---

- **What it measures**

- VO<sub>2</sub>max
- Moderate-vigorous min/wk
- Daily steps

- **Reservoir / coupling**

Cardiorespiratory reserve (an asset that protects CV & MET).

- **Why it matters**

VO<sub>2</sub>max is a top mortality predictor — a  $\beta$ -blocker confounds it (down-weighted).





# Mental / Cognitive

Mood, stress, cognition & connection.

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- **What it measures**

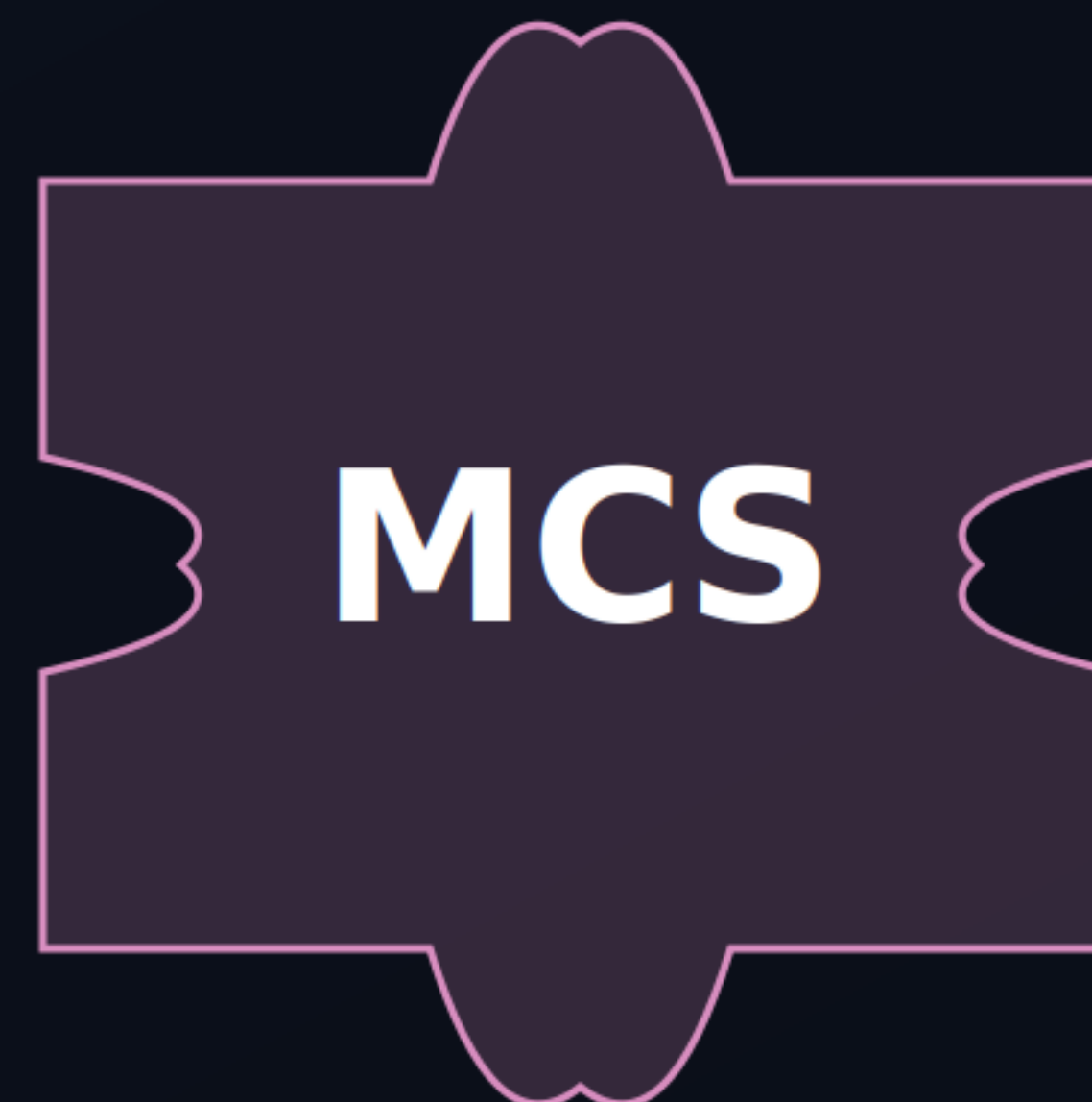
- PHQ-9 (depression)
- GAD-7 (anxiety)
- Loneliness (UCLA-3)

- **Reservoir / coupling**

Allostatic load couples to sleep, endocrine & CV.

- **Why it matters**

A hard safety rule: self-harm flag forces immediate crisis escalation.



## SUMMARY

# One score.

## Twelve pieces. A trust layer. A loop.

- **Worst-sensitive score that can't hide a crisis**
- **Early-warning that catches disease incubating**
- **Context-aware & UAE-localized, evidence-cited**
- **Clinician-in-loop, equity-firewalled, auditable**
- **Compounds with data — the AI doctor that improves**

Built on the Babylon discipline: no claim ships ahead of its evidence.

**Today: a defensible design. Next: the data that proves it.**

